Dave’s Daily

tldr

1. On Monday Homeland Security posted an order that civilian agencies remove DeepSeek and stop buying it, including tools built with its models. Bloomberg dates the signed order 24 July. On an outside index the 17 October model scores 45, second of 121.

2. On Monday S&P’s Mark Habib said hyperscalers likely owe more for AI than their disclosures show. Six firms are expected to spend more than $7 trillion from 2025 through 2030, with negative free cash flow until at least next year.

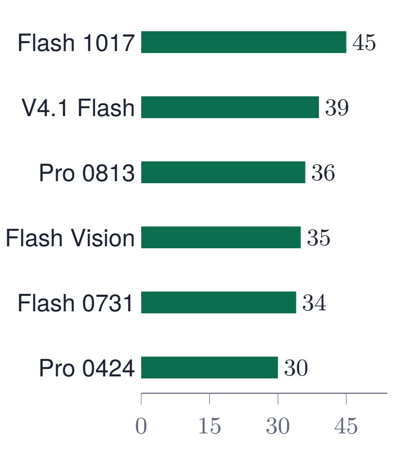
3. A 17 October paper cancels noise with a cheap paired measurement. The cheap average need not resemble the objective. At squared correlation 0.8, cumulative regret after 1,000 trials fell 34.8 to 50.6 percent versus ordinary GP-UCB.

AI

On Monday the Department of Homeland Security posted a public notice, scheduled for the Federal Register on Tuesday, that civilian federal agencies must remove DeepSeek products and stop buying them. Bloomberg reports that the signed order is dated 24 July and is signed by Secretary Markwayne Mullin. The BBC reports that the ban took effect from the end of July, that it covers technology built with DeepSeek’s models, and that the Defense Department is outside it. Treasury, State, and Justice are among the civilian departments inside it.

CyberScoop quotes Mullin on the reasons. DeepSeek stores data on servers in China, where national-security law lets the government request that data. Models hosted by DeepSeek collect personal information and a record of online activity. A prompt and a reply can carry sensitive material, and Mullin writes that the combination with public data could produce intelligence on U.S. persons and government programs. CyberScoop notes that the prompt concern would apply to other models as well. An agency may ask for an exception. The request has to arrive 60 days ahead and has to give a compelling justification, such as a mission the agency cannot otherwise perform.

The order binds those agencies. It does not bind a private firm. The BBC reports that Snowflake, Vercel, and OpenRouter still include DeepSeek in products they sell. When a civilian agency uses one of those products, the removal covers that use.



Artificial Analysis Intelligence Index for six DeepSeek Max (reasoning) releases. Longer is a higher score. The axis starts at zero. Flash 1017, released 17 October, scores 45 and ranks 2 of 121 models on that index. The median among comparable models is 19. Source: Artificial Analysis, read 20 October 2026.

The same page prices a Flash 1017 task on that index at $0.26, with the API at $0.30 per million input tokens and $1.20 per million output tokens. The score of 45 is six points above the prior V4.1 Flash release, which scores 39, and nine points above the 13 August Pro release, which scores 36 with a larger model. Civilian agencies are being told to remove a model family that, on this outside index, now sits near the top.

Markets

On Monday in London, S&P Global’s head of digital infrastructure, Mark Habib, said the largest technology firms are likely to owe more for AI than their disclosures show. Marc Jones, writing for Reuters, reports the comments from a roundtable. The six firms in S&P’s total are Amazon, Microsoft, Alphabet, Oracle, Meta, and SpaceX. S&P expects them to spend more than $7 trillion on data centers and other AI capital projects from 2025 through 2030, and to have negative free cash flow until at least next year.

Habib calls the missing piece interconnected financing: a large firm guarantees a data-center build, or a smaller AI company. If the guarantee is absent from the firm’s disclosure, and if S&P is not rating that transaction, S&P does not see the liability. Habib said, “It is safe to assume that there is more (debt) out there that we are not seeing.” He gave no figure for the extra debt. S&P is not changing ratings on that gap. Habib said the larger risk to the ratings is that the expected profits do not arrive. In July S&P cut Oracle to BBB-, its lowest investment-grade rating, citing AI spending and what S&P called an uncertain path to profitability. A change in the present view, Habib said, would require a judgment that the spending is no longer sustainable, from still larger investments or from competition that presses revenue hard. Reuters also notes that bonds of AI infrastructure firms, including CoreWeave, have been trading at stressed prices. The new claim is the debt S&P cannot see.

Bayesian optimization

arXiv:2610.24737, submitted 17 October, is *Gaussian Process Bandits with Paired Auxiliary Observations*, by Weibo Huang and Cheng Hua (Shanghai Jiao Tong University) and Qing Feng and Ruihao Zhu (Cornell).

A trial returns a noisy score, and the noise makes the Gaussian process learn more slowly. The same trial often also returns a cheap second number from the same random draw: a low-fidelity simulator, a sensor, or a model score. AI-GPB uses the pair to shrink the noise. The cheap number’s average does not have to resemble the score being optimized, and it does not have to share the same best point.

Before the search, the method fits that cheap average from cheap data alone. During the search, each trial returns both numbers. One coefficient, learned from the pairs, says how much of the expensive noise moves with the cheap noise. The method subtracts that shared part and gives the corrected score to ordinary GP-UCB. The next query can move because the posterior is tighter.

The experiments build the cheap average from a surface that does not depend on the objective, so a gain is not a copy of the answer. Table 1 gives cumulative regret after 1,000 trials, a mean over 20 runs, on four two-dimensional functions. At a squared correlation of 0.8 between the paired errors, AI-GPB cuts that regret by 50.6 percent on Ackley, 40.3 percent on Branin, 40.6 percent on Rosenbrock, and 34.8 percent on the six-hump camel, relative to GP-UCB. At a squared correlation of 0.25 the cuts are 8.7, 9.2, 14.3, and 6.6 percent, in that same order.

Simple regret is the best point found, rather than the whole path. After 300 expensive trials, at the same squared correlation of 0.8, geometric-mean simple regret is between 22.7 and 72.4 percent lower than GP-UCB. That comparison spans problems from 2 to 10 dimensions. On five of the six tasks, the result finishes within 17.3 percent of a reference that knows the true cheap average and the best coefficient. Hartmann, in six dimensions, is the task with the larger gap. With 128 offline locations and squared correlation 0.8, mean cumulative regret on Ackley falls by about 43 percent, and the four two-dimensional tasks fall by 35 to 50 percent.

The gain tracks the correlation. If the paired errors do not move together, little noise remains to cancel. The paper’s Matérn regret bounds shrink as that leftover noise shrinks.

Updated 20 October 2026. Phone and letter match. Sources: Bloomberg, the BBC, and CyberScoop, 19 October; the Federal Register notice scheduled for 20 October; Artificial Analysis, read 20 October; Reuters, 19 October; arXiv, 17 October. The removal order is a legal instruction to civilian agencies, not a finding that a particular prompt was leaked. The $7 trillion figure is S&P’s spending total, not a debt total. The regret cuts are means over the paper’s runs. Created with malvin.